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(54) **RETURN CHANNEL FOR A SURF POOL**

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E04H 4/00 (2006.01)
A63B 69/00 (2006.01)

(52) **U.S. Cl.**
CPC ***E04H 4/0006*** (2013.01); *A63B 69/0093* (2013.01)

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CPC E04H 4/0006; E04H 4/00; A63B 69/0093; A63B 69/125; B63B 34/70
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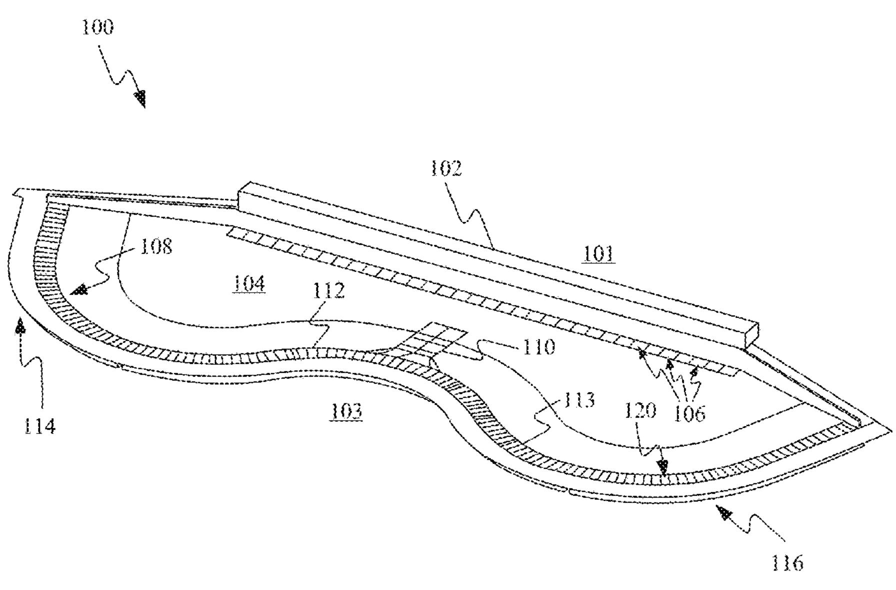
(57) **ABSTRACT**

A return channel collects and redirects water back into a surf basin of a surf pool attraction. The return channel can be covered by grating panels that improve navigability of the surf pool attraction while still allowing water to be collected in the return channel. The return channel can include a peripheral channel portion that extends around a periphery of one side of the surf pool and a central channel portion that connects to and extends from the peripheral channel portion toward a central region of the surf basin.

23 Claims, 8 Drawing Sheets



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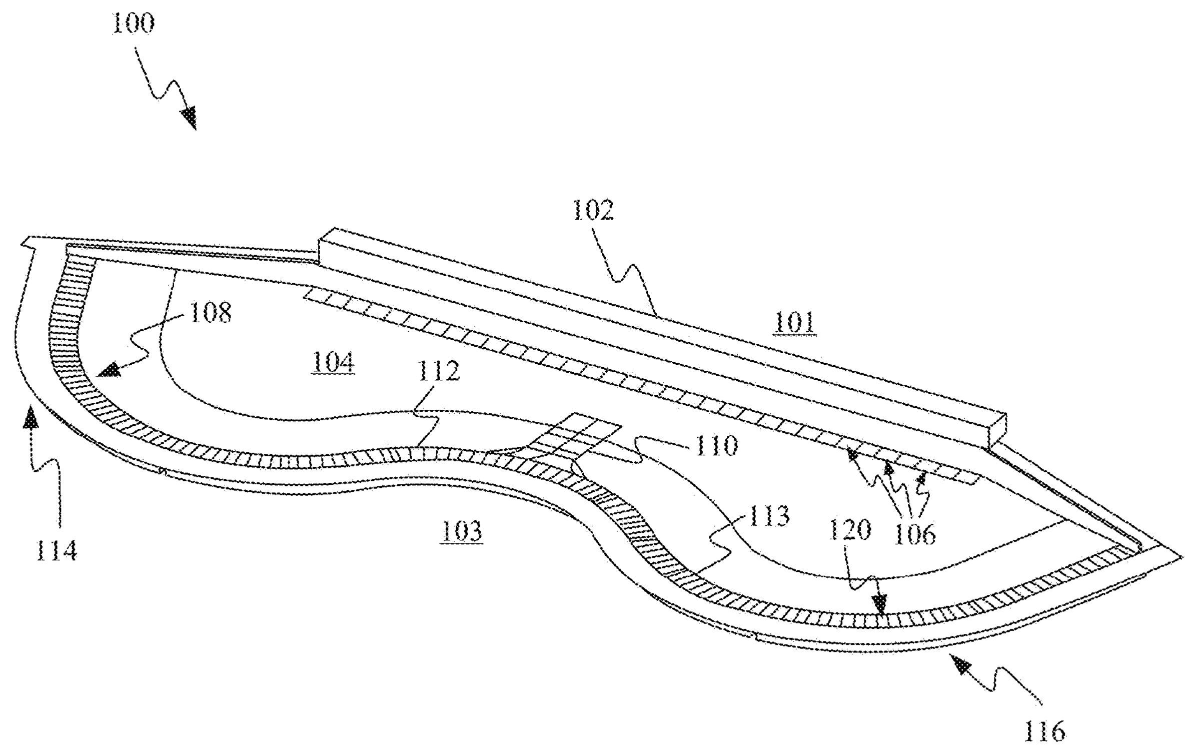

FIG. 1

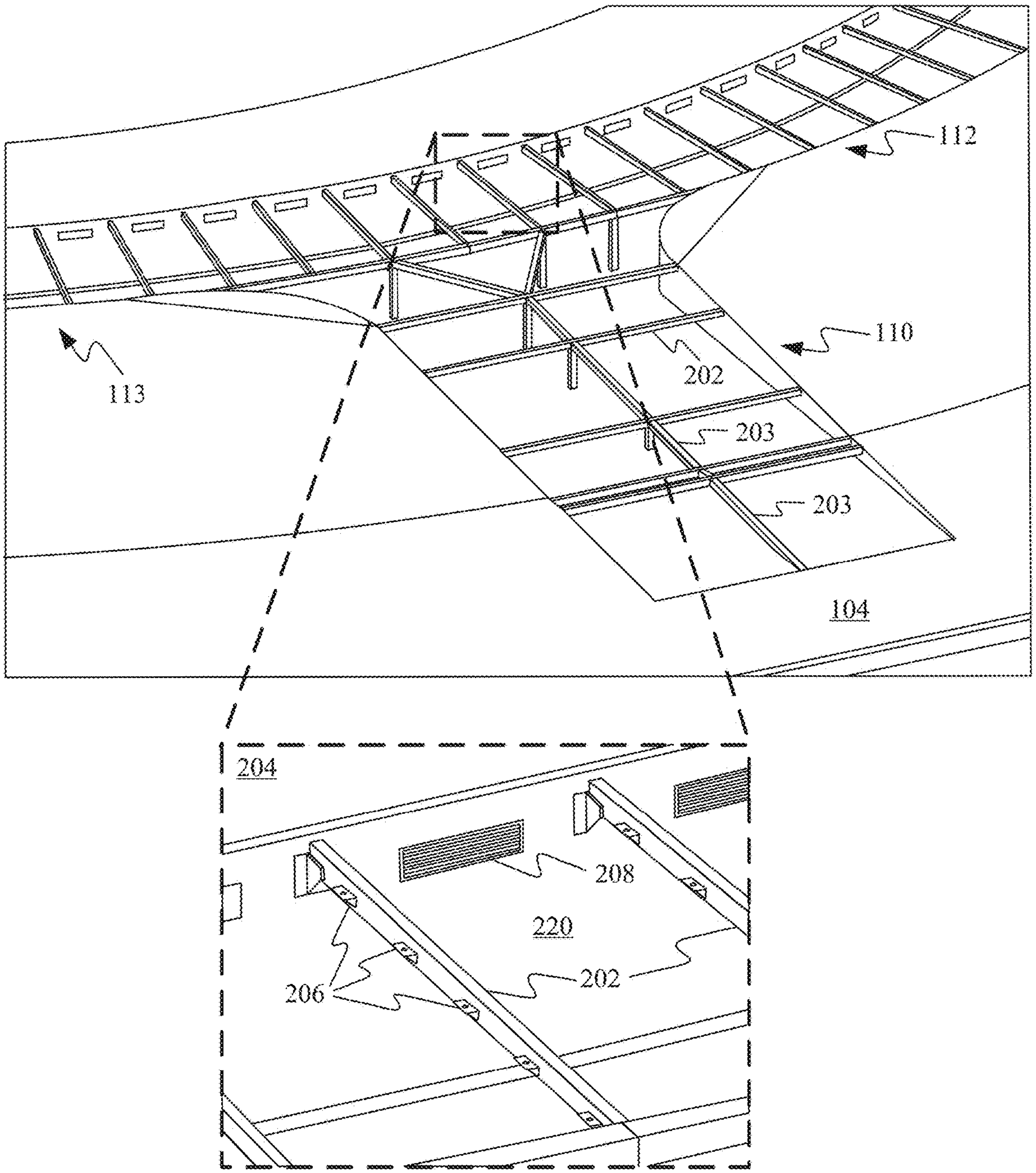

FIG. 2

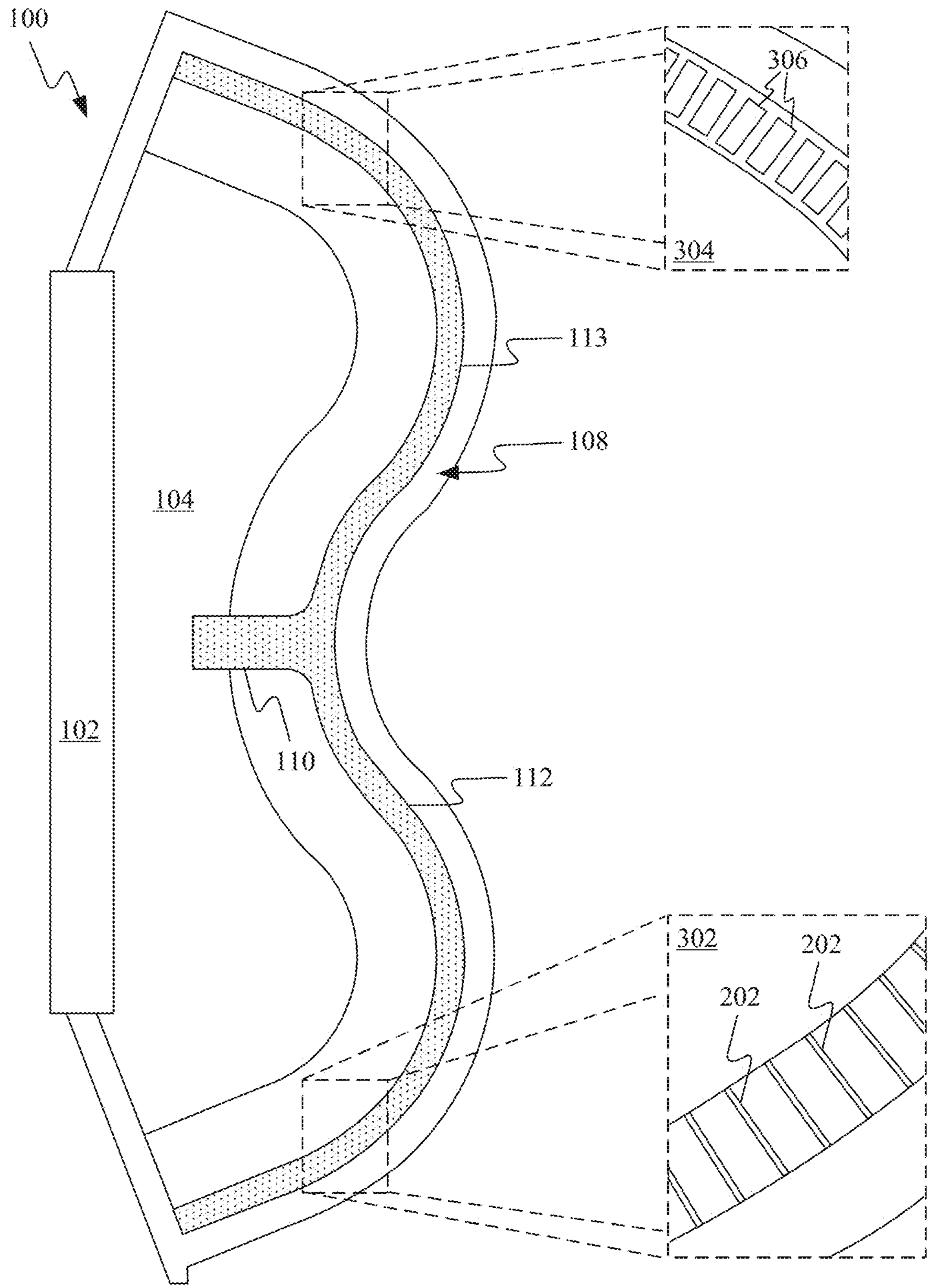

FIG. 3A

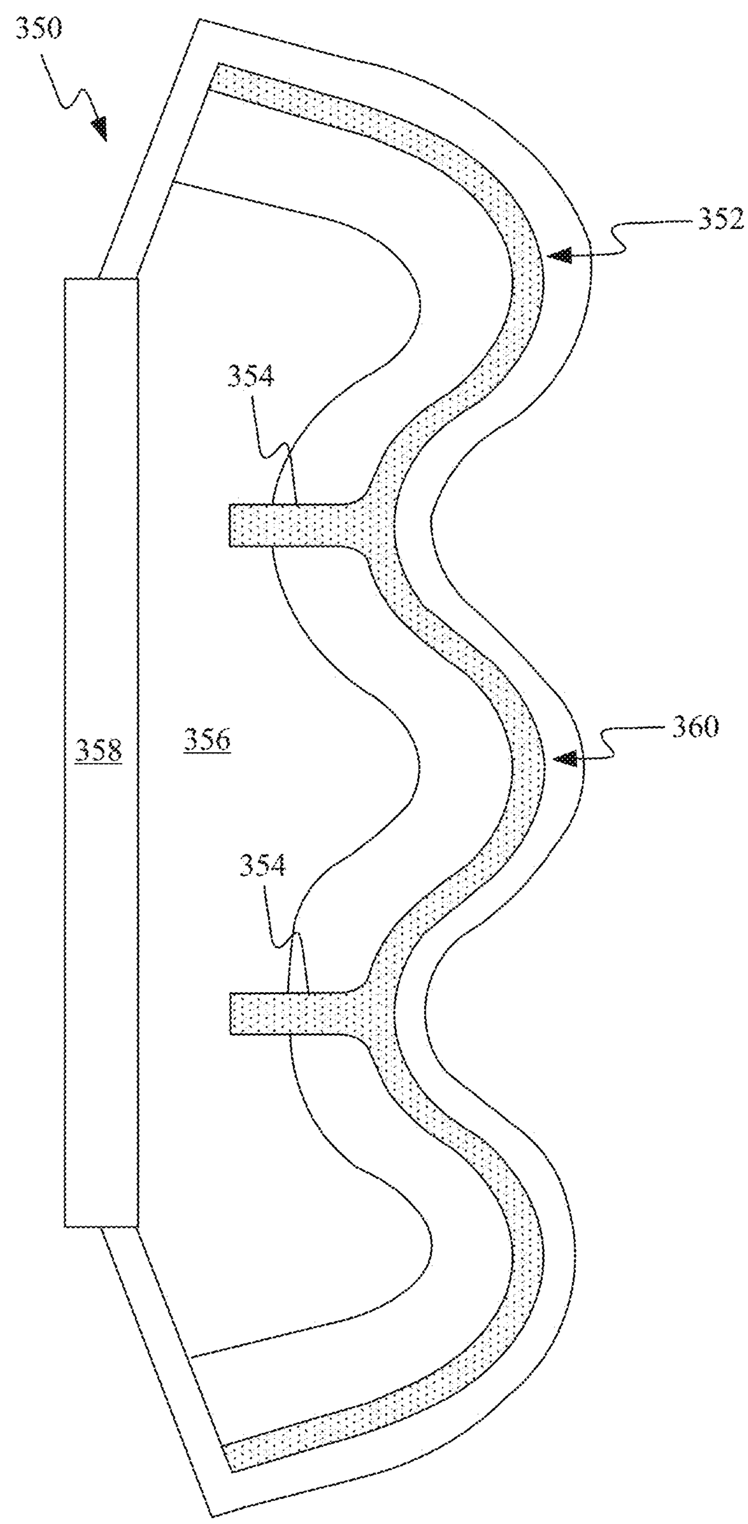

FIG. 3B

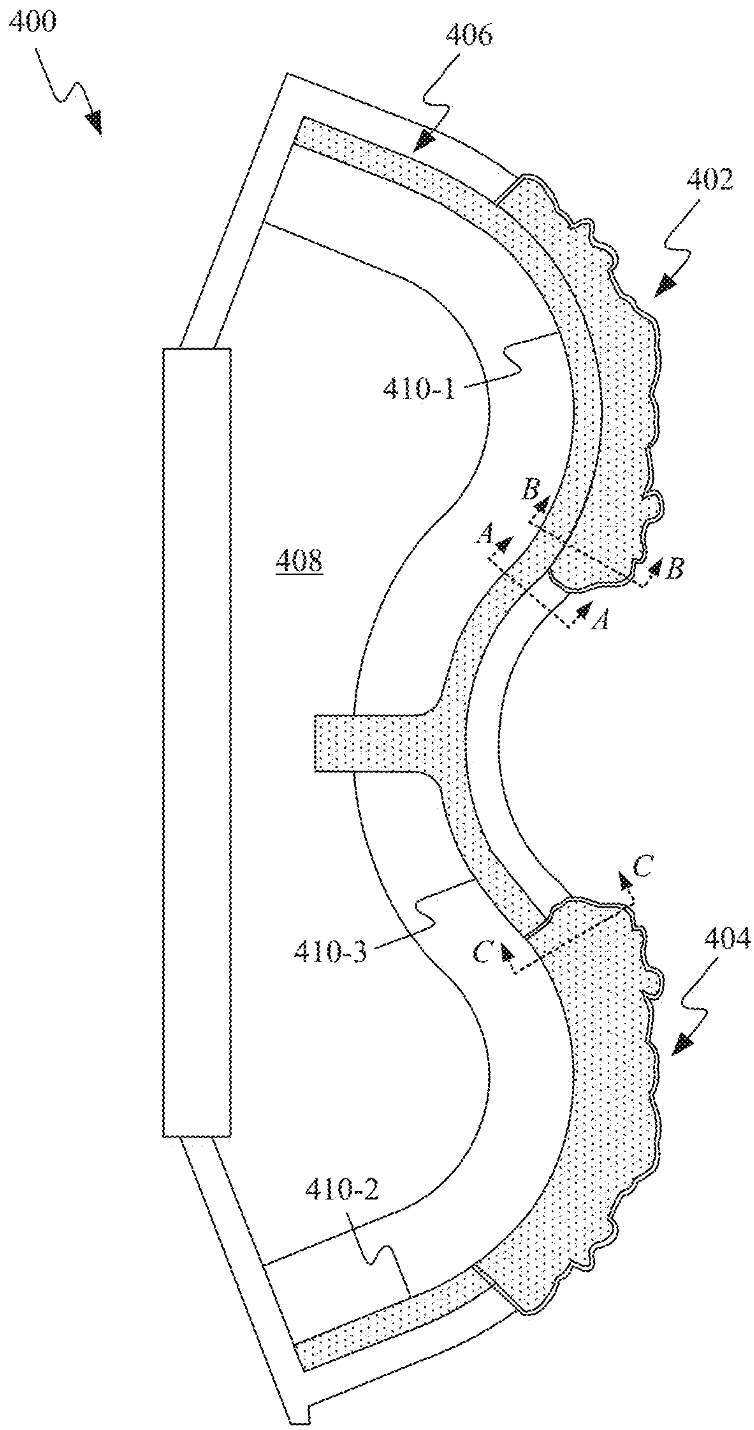

FIG. 4A

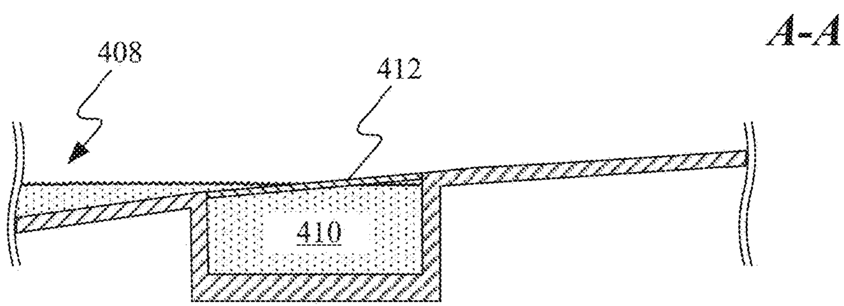

FIG. 4B

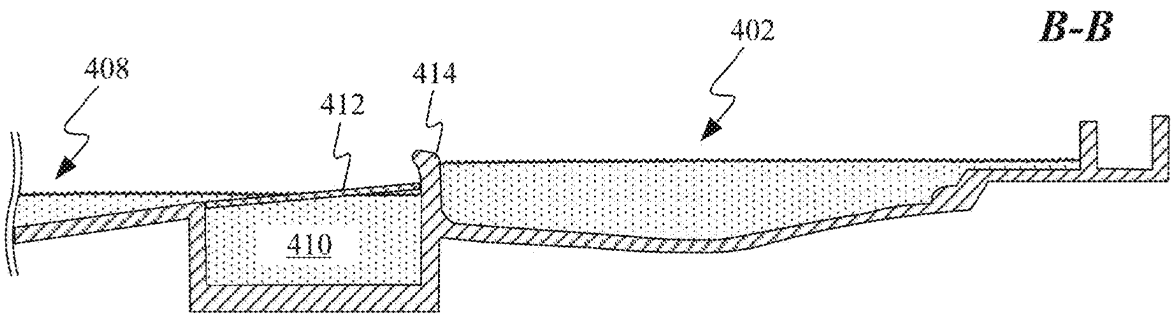

FIG. 4C

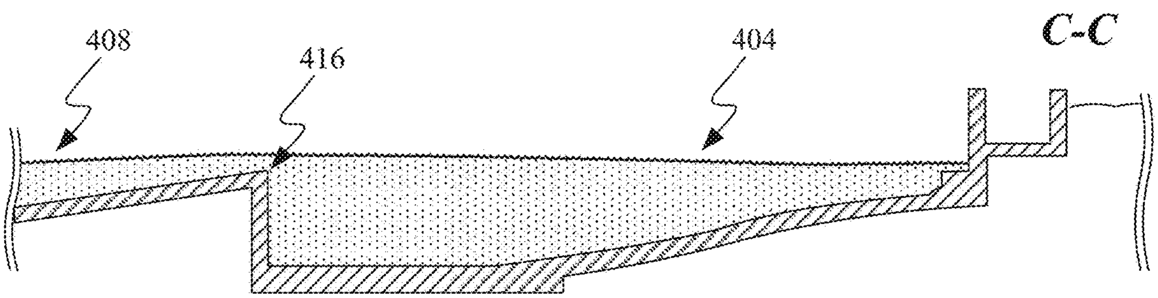

FIG. 4D

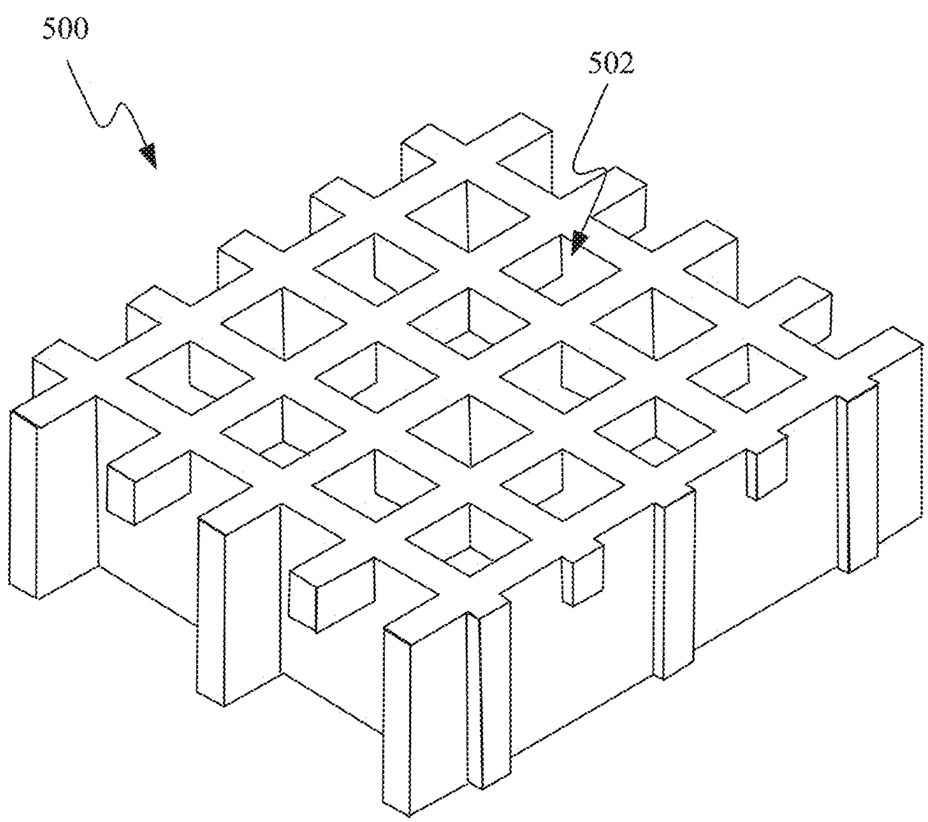

FIG. 5

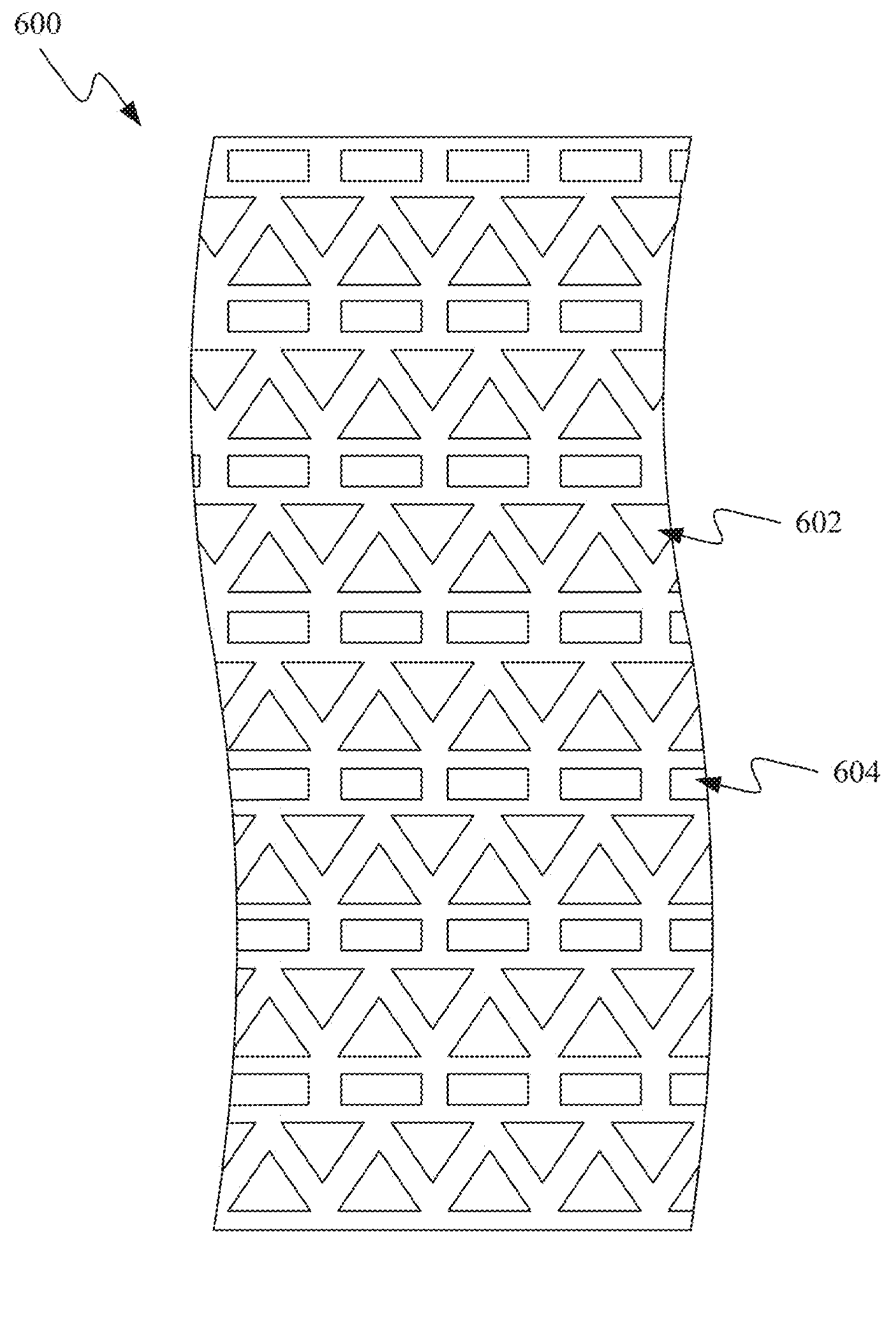

FIG. 6

RETURN CHANNEL FOR A SURF POOL

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of, and is a continuation in part of U.S. Non-Provisional patent application Ser. No. 18/057,150. U.S. Non-Provisional patent application Ser. No. 18/057,150 is a national stage entry of PCT Application No. PCT/CA2021/000045, filed May 18, 2021, which claims priority to U.S. Provisional Patent Application No. 63/026,508, filed May 18, 2020, all of which are incorporated by reference in their entirety and for all purposes.

TECHNICAL FIELD

The embodiments discussed in the present disclosure are related to various configurations for a channel configured to return water back into a surf or wave pool.

BACKGROUND

Water parks often include a variety of water related attractions. Surf pools are one of the many water related attractions that can be provided. Surf pools invariably include movement of large volumes of water to create a series of waves configured for surfing enthusiasts. The waves generated often move at least some of the water to a periphery of the surf pool. Extending the size of the surf pool water attractions to collect the water often unnecessarily extends the size of the surf pool attraction, taking up space that can often be used more efficiently for other attractions. For this reason, solutions that allow space to be saved without sacrificing aesthetics or utility of the surf pool are desirable.

The subject matter claimed in the present disclosure is not limited to embodiments that solve any disadvantages or that operate only in environments such as those described above. Rather, this background is only provided to illustrate one example area where some embodiments described in the present disclosure may be practiced.

SUMMARY

Disclosed are various embodiments that relate to managing the movement of large volumes of water and water channel configurations operable to efficiently return water to a surf pool.

A surf pool is described and includes a surf basin; a wave generator disposed on a first side of the surf basin that is configured to generate waves that travel from the first side of the surf basin to a second side of the surf basin opposite the first side of the surf basin; and a return channel extending along at least a portion of a periphery of the second side of the surf basin. In some embodiments, the return channel includes a central channel that extends from the periphery of the second side of the surf basin and into a central portion of the surf basin.

In some embodiments, multiple edge pools can be arranged behind and in close proximity to the return channel. Various configurations are possible including some where the edge pool is either incorporated into the return channel or separated from the return channel by a common wall.

A return channel is described that includes a peripheral channel portion extending along a periphery of a first side of a surf basin; a central channel portion that is connected to and protrudes from the peripheral channel portion and into a central region of the surf basin; and a plurality of grating panels covering the peripheral channel portion and the central channel portion.

Other aspects and advantages of the described embodiments will become apparent from the following detailed description taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the described embodiments.

INCORPORATION BY REFERENCE

All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

U.S. Pat. No. 11,891,834 issued Feb. 6, 2024 for Chamber and Control System and Method for Generating Waves.

BRIEF DESCRIPTION OF THE DRAWINGS

Example embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

FIG. **1** shows a perspective view of an exemplary surf pool;

FIG. **2** shows a perspective view of a central region of a return channel of the surf pool depicted in FIG. **1**;

FIG. **3**A shows a top-down view of the exemplary surf pool depicted in FIG. **1** with close up views of different portions of the return channel;

FIG. **3**B shows a top-down view of another surf pool and an overall shape of another return channel;

FIG. **4**A shows a top-down view of another exemplary surf pool with multiple edge pools arranged in close proximity or integrated with the return channel;

FIG. **4**B shows a cross-sectional side view of the surf pool depicted in FIG. **4**A in accordance with section line A-A from FIG. **4**A;

FIG. **4**C shows a cross-sectional side view of the surf pool depicted in FIG. **4**A in accordance with section line B-B from FIG. **4**A;

FIG. **4**D shows a cross-sectional side view of the surf pool depicted in FIG. **4**A in accordance with section line C-C as shown in FIG. **4**A; and

FIG. **5** shows a perspective view of an exemplary section of a grating panel in accordance with the described embodiments.

FIG. **6** shows a top-down view of another exemplary section that can be incorporated into the grating panel shown in FIGS. **4**A-**4**D and includes different shaped openings

DETAILED DESCRIPTION

A surf pool utilizes a device often referred to as a wave generator to create a series of surf waves. Wave generators can be configured to produce a variety of different types of waves with varying sizes and shapes suitable for surfers of various experience and skill level. Accordingly, the surf pool should be operable to provide customizable waves for multiple ability levels.

FIG. **1** shows an exemplary surf pool **100** in accordance with the disclosure. Surf pool **100** includes a wave generator assembly **102** located on a first side **101** of surf pool **100**. The wave generator assembly **102** is operable to generate waves that travel from the first side **101** of surf pool **100** to a second side **103** of surf pool **100**. In some embodiments, wave generator assembly **102** can take the form of a pneumatically driven wave generator assembly. When wave generator assembly **102** takes the form of a pneumatically-driven wave generator assembly, the wave generator assembly **102** includes multiple chambers operable to be filled with pressurized air, which in turn periodically pushes water into a surf basin **104** through one or more delivery channels **106** in a manner suitable for generating waves. It should be appreciated that while a specific wave generator assembly **102** is depicted and described, the described embodiments are compatible with a wide variety of wave generator assemblies. For example, FIG. **1** shows a configuration with 34 delivery channels **106**. As will be appreciated by those skilled in the art, a larger or smaller number of delivery channels could be used depending on the size of a respective surf pool and an amount of flexibility desired for the types of waves to be generated in the surf pool **100** without departing from the scope of the disclosure. For example, configurations with between 10 and 100 delivery channels are also possible and believed to be within the scope of the described embodiments.

A return channel **108** is positioned at the second side **103** of surf pool **100** and operable to capture portions of the generated waves as they approach a periphery of surf pool **100**. The periphery is the edge of the surf pool **100** opposite the delivery channels **106**. Return channel **108** can be sized to largely prevent water from inadvertently escaping surf pool **100** during operation of wave generator assembly **102**. Once water is captured within return channel **108**, the water can be returned into a central portion of surf basin **104**. In some embodiments, return channel **108** can be covered by grating panels with an upward facing surface that allows guests or users of the surf pool **100** to conveniently walk over any portion of return channel **108** while still accommodating the passage of water into the return channel **108**. Alternatively, guests or users can move over an uncovered portion or portions of return channel **108** using one or more bridges that span return channel **108**. Some configurations can include portions of return channel **108** covered by grating panels and other portions remaining uncovered. Bridges or narrow walkways could be operable to allow for passage over any portions of return channel **108** not covered by grating.

A width of return channel **108** can vary. However, widths of between two and six meters allow for large volumes of water to be captured when wave generator assembly **102** is creating its largest waves. The width of return channel depicted in FIG. **1** is about four meters and its depth fixed at about 1.5 meters in one particular embodiment. As will be appreciated by those skilled in the art, the depth of the channel **108** can be greater or less than 1.5 meters and can vary in depth along the length of the channel. For example, in one embodiment a depth of the channel can gradually increase towards the center to encourage larger volumes of water to return more quickly to the surf basin **104**. In another embodiment, the depth of the channel can decrease towards the center to encourage water to move toward the ends of return channel **108** initially.

As depicted, return channel **108** is positionable along all or part of the second side **103** of the surf pool **100** opposite the delivery channels **106**. The return channel **108** can be configured to have a first side and a second side with a central channel portion **110** that protrudes into a central region of surf basin **104**, a first peripheral channel portion **112** that extends away from the central channel portion **110** in a first direction and a second peripheral channel portion **113** that extends away from the central channel portion **110** in a second direction opposite the first direction. The first and second peripheral channel portions **112** and **113** follow a periphery of the second side **103** of surf pool **100**. In some embodiments, one or more areas of sand can be positioned 5-10 m past return channel **108** in order to form a beach where guests can congregate. The beach area can also be operable to soak up water and dissipate the force of any waves that end up extending past return channel **108**. This can be helpful in scenarios where a section or sections of return channel **108** are flooded with water.

Water captured in the peripheral channel portions **112** and **113** is able to return to surf basin **104** by way of central channel portion **110**. In some embodiments, central channel portion **110** can be omitted and water captured within return channel **108** can be returned into surf basin **104** using one or more water pumps. For example, instead of having the depicted central channel portion **110**, one or more large intake pipes could be provided. The intake pipes can then be configured to carry water retained by return channel **108** away for redistribution of the water throughout the surf basin **104** using a series of outlet valves.

Central channel portion **110** can also be covered by the same or a similar grating panels **120** overlaying peripheral channel portions **112** and **113**. Covering central channel portion **110** with a grating helps avoid discontinuities in the slope and surface of surf basin **104**. However, in addition to providing a consistent surface and an efficient mechanism for returning water into surf basin **104**, the grating covering central channel portion **110** by virtue of its hole spacing and position within surf basin **104** is operable to dissipate energy in a central portion of surf basin **104** that reduces eddies and other undesired currents resulting from generation of waves by wave generator **102**. Reducing eddies and other undesired currents can be particularly helpful in situations where wave generator **102** is concurrently producing a first set of waves travelling toward a first end **114** of surf pool **100** and a second set of waves travelling toward a second end **116** of surf pool **100**. In some embodiments, this can allow wave generator assembly **102** to produce a more consistent set of waves without having to wait as long for the surface of the water within surf basin **104** to calm between sets of waves.

FIG. **2** shows a detailed perspective view of the central channel portion **110** and how central channel portion **110** is connected to peripheral channel portions **112** and **113** of return channel **108**. Grating panels have been removed from the view shown in FIG. **2** in order to illustrate a configuration of a grating support structure. The grating support structure is operable to support and maintain a position of one or more grating panels over return channel **108**. When the grating panel is firmly supported, the grating panel does not change orientation when the weight of a user is applied at any location along the grating panel or when waves impact the grating panels. The grating panels are operable to cover the return channel such that one or more of the plurality of grating panels are in direct contact with the wall shared by the return channel and the edge pool. As depicted, the grating support structure includes multiple beams **202** extending across a width of return channel **108**. A space **220** is provided between adjacent beams **202**. The space **220** between adjacent beams **202** can be sized to accommodate a grating panel sized to be positioned at a location on the grating support structure so that opposing sides of the grating panel are firmly supported. As depicted, the central channel portion **110** of the return channel **108** is about twice the width of, e.g., first peripheral channel portion **112**. In some embodiments, this allows for grating panels used for the peripheral channel portion **112** to be interchangeable with grating panels used for the central channel portion **110**. This illustrated configuration is achieved here by arranging one or more central beams **203** within the central channel portion **110** along a centerline of central channel portion **110**.

FIG. **2** also includes a close-up view **204** that shows additional detail for some of the beams **202** disposed within peripheral channel portion **112**. In some embodiments, beams **202** can take the form of steel beams that include multiple brackets **206** protruding laterally from either side of a respective beam **202** to allow for convenient attachment of the grating panels to beams **202**. For example a fastener can be driven through a first fastener opening in a grating panel and then engage a second fastener opening in a bracket **206** to secure the grating panel in place. Close-up view **204** also shows a drain **208** near a top portion of peripheral channel portion **112** that prevents or mitigates situations in which peripheral channel portion **112** overfills due to rain or other circumstances.

FIG. **3**A shows a top-down view of surf pool **100** and an overall shape of return channel **108**. Return channel **108** is emphasized here using a dot pattern. Central channel portion **110** is shown extending into a central region of surf basin **104**. While the serpentine geometry provides a pleasing aesthetic appearance, the concave curvature of the peripheral portion of peripheral channel portion **112** also allows a length of central channel portion **110** to be shorter in length than it would otherwise be in the event return channel had an entirely convex curvature.

Close-up view **302** shows a top view of a configuration in which grating panels are supported by beams **202** as shown in FIG. **2**. Close up view **304** shows an alternative support structure configuration for the grating panels using multiple concrete piers **304**. Concrete piers **304** can extend from a bottom surface of peripheral channel portion **112** close to a top of peripheral channel portion **112** to support grating panels covering the top of peripheral channel portion **112**. In a configuration using concrete piers **304** peripheral channel portion **112** can be slightly deeper to account for space within return channel **108** taken by concrete piers **304**. In some embodiments, a mix of beams **202** and concrete piers **306** can be used to support grating panels covering return channel **108**. In other embodiments, either beams **202** or concrete piers **306** are used to support grating panels extending over the entirety of return channel **208**.

FIG. **3**B shows a top-down view of a surf pool **350** and an overall shape of return channel **352**. Return channel **352** serves a similar purpose to previously depicted return channel **108** but includes two central channel portions **354** extending into surf basin **356**. This configuration can be beneficial in cases where wave generator assembly **358** is configured to send different types of waves to three distinct sections of surf pool **350**.

Furthermore, in some embodiments, return channel **352** can be divided into two separate return channels. For example, return channel **352** can be divided into multiple return channels by dividing return channel **352** at point **360**, allowing for independent return channels to feed each of central channel portions **356**. Any of the surf pools depicted herein could include return channel configurations with multiple return channels arranged to follow the periphery of the surf pool or in other embodiments, discrete rows of return channels could be arranged to help collect water and/or dissipate wave energy.

FIG. **4**A shows a top-down view of a surf pool **400** similar in operation to surf pool **100** that includes multiple integral edge pools (e.g. first edge pool **402** and second edge pool **404**) formed adjacent to or integrated with a return channel **406** configured to redirect water back into surf basin **408**. While this particular surf pool **400** includes only two distinct edge pools, it should be appreciated that a surf pool could be configured with a smaller or larger number of edge pools. For example, one, three, four or five edge pools are also possible. Also smaller pools can be formed within the bounds of integral edge pools **402** and/or **404**. Edge pool **402** shares a wall with a peripheral channel portion **410-1** of the return channel **406**. Edge pool **404** extends into and allows a peripheral barrier associated with a section of peripheral channel portion **410** to be removed so that edge pool **404** can incorporate space otherwise taken by peripheral channel portion **410**. The edge pool configurations can be a great choice when more variability is desired within a lounging pool. These configuration allow for some disturbances of the water. For example, some disturbances would result from waves sometimes flowing into and out of edge pool **404**. It should be noted that while edge pool **404** is shown with a barrier separating it from adjacent sections of the return channel **406** that the barrier can be entirely removed to allow edge pool **404** to be completely incorporated into the return channel **406**

In some embodiments, water within different sections of peripheral channel portion **408** is in fluid communication with and therefore allowed to freely circulate between edge pool **404** and adjacent sections of peripheral channel portion **410**. Additionally, edge pool **404** can include a series of valves and/or pumps that help to maintain a desired water level within edge pool **404**. The use of pumps and valves can also be operable to help regulate a level of water in the adjacent sections of peripheral channel portion **410**. For example, edge pool **404** can act as a pass through that allows water from section **410-2** of peripheral channel portion **410** to move into peripheral channel portion **410-3** in the event water levels in peripheral channel portion **410-2** exceed a predetermined threshold level or in the event of a threshold height differential in the water contained within sections **410-2** and **410-3** of peripheral channel portion **410**. Alternatively, a floor of peripheral channel portion **410-2** can be higher than a floor of peripheral channel portion **410-3** and piping can be arranged to transfer water slowly from peripheral channel portion **410-2**, through edge pool **404** and to peripheral channel portion **410-3**. In some embodiments, this piping can be arranged along the floor or just beneath the floor of edge pool **404**.

FIG. **4**B shows a cross-sectional side view of surf pool **400** in accordance with section line A-A as shown in FIG. **4**A. In particular, FIG. **4**B shows a cross-sectional view of peripheral channel portion **410** covered by grating panel **412** and how grating panel accommodates the passage of water into peripheral channel portion **410** while also allowing for convenient movement of a user over peripheral channel portion **410**. As depicted, an inclined slope remains substantially similar to the slope of the terrain leading out of surf basin **408**, thereby lowering the risk of guests tripping or losing their balance as they cross peripheral channel portion **410**. In some embodiments, grating panel **412** can be installed and arranged such that the upward facing surface of the grating panel is substantially flush with an upward facing surface of surf basin **408** and such that the upward facing surface of the grating panel follows an upward slope of the surf basin. While the cross-sectional views show grating panel following a slope of the wave pool, it should be appreciated that in some embodiments a slope of the grating can increase in order to orient the grating more directly at incoming waves and thereby capture more water. It should be noted that the grating panels used to cover the return channel of surf pool **100** can have a similar arrangement to the grating panel **412** as depicted here in FIG. **4**B. Moreover, the grating panels are not limited to planar configurations (e.g., horizontally planar). The grating panels can have a contoured upper surface (e.g., slightly concave or convex) as a result of the contouring and shaping of the surf basin **408** of the surf pool at any particular location adjacent the return channel.

FIG. **4**C shows a cross-sectional side view of surf pool **400** in accordance with section line B-B as shown in FIG. **4**A that shows peripheral channel portion **410** and edge pool **402**. As depicted, water positioned within edge pool **402** and peripheral channel portion **410** is separated by a common channel wall **414**. In some embodiments, a distal end of channel wall **414** can be decorated with rocks or other barriers that help reduce an amount of water splashing into edge pool **402** as a result of waves being generated within surf basin **408**. As depicted, the presence of sloped grating panel **412** helps guests to more easily enter and exit edge pool **402** from surf basin **408** by reducing an apparent height of channel wall **414** with a guest standing upon a portion of grating panel **412** proximate channel wall **414**.

FIG. **4**D shows a cross-sectional side view of surf pool **400** in accordance with section line C-C as shown in FIG. **4**A that shows peripheral channel portion **410** and edge pool **404**. As depicted water from surf basin **408** is able to splash freely into edge pool **404**, which acts as extended storage for return channel **410** by collecting some waves that send water to the periphery of surf pool **400**. Between waves water levels can sink just below an outer lip **416** of surf basin **408**. As mentioned previously, the illustrated configuration is operable to allow the space generally taken up by peripheral channel portion **410** to contribute to a size of edge pool **404**. This configuration also removes the need for an outer channel wall, similar to channel wall **414** shown in FIG. **4**C. In some embodiments, edge pool **404** can be open to peripheral channel portion **410**. In some embodiments and as depicted in FIG. **4**A, edge pool **404** can be walled off from adjacent sections **410-2** and **410-3** of peripheral channel portion **410**. Similar to the embodiments described in conjunction with edge pool **402**, edge pool **404** can also be configured to allow for the passage of water between sections **410-2** and **410-3** or peripheral channel portion **410** as needed. Walling off edge pool **404** from sections **410-2** and **410-3** can allow for more control over the water levels within edge pool **404**. FIG. **4**D shows how simple it is for a guest to transition from surf basin **408** to edge pool **404** simply by stepping past the outer lip **416** of surf basin **408**.

FIG. **5** shows a perspective view of an exemplary section **500** that can be incorporated into the grating panel **412** as shown in FIGS. **4**A-**4**D. An actual grating panel **412** would be much larger than depicted and in some embodiments have a size of about one meter by four meters. Additionally, the upper surface of the grating panel can be planar as illustrated or have or achieve contouring. The shape in a first plan can be square or rectangular. In some configurations, one or more sides of the grating panel can be curved. Openings **502**, can be square, as illustrated, or any other suitable shape including, but not limited to rectangular, triangular, round, oval, and ovoid. In some embodiments, a size of openings **502** can be about 8 mm square in a first dimension. The dimensional size of openings **502** of the grating can be varied for different effects. For example, grating openings **502** having smaller sizes, e.g. between 4 mm and 8 mm can also be used and are considered within the scope of the described embodiments. A thickness of grating panel **412** can vary between 20 and 80 mm in thickness and be formed from a polyester resin in some embodiments.

FIG. **6** shows a top-down view of another exemplary section **600** that can be incorporated into the grating panel **412** as shown in FIGS. **4**A-**4**D and includes different shaped openings. In particular, section **600** includes triangular openings **602** and rectangular openings **604**, illustrating how a single section of a grating panel can include multiple differently shaped openings. Grating panels can include openings of different shape and size to fine tune a desired amount of energy dissipation and wave pass through for a particular surf pool configuration.

As described earlier, the interaction between the water driven by wave generator assembly **102** and the grating over central channel portion **110** is effective at dissipating energy from the waves. When water is reentering the surf basin through central channel portion **110**, testing showed more than a 60% reduction in velocity for water passing through a grating panel having an 8 mm square grating pattern. Testing also showed the interaction of the waves with the grating having a smoothing effect on the waves. In some embodiments, a slope of the central channel portion of the return channel can vary between about four and six degrees.

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter configured in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

Unless specific arrangements described herein are mutually exclusive with one another, the various implementations described herein can be combined in whole or in part to enhance system functionality and/or to produce complementary functions. Likewise, aspects of the implementations may be implemented in standalone arrangements. Thus, the above description has been given by way of example only and modification in detail may be made within the scope of the present invention.

With respect to the use of substantially any plural or singular terms herein, those having skill in the art can translate from the plural to the singular or from the singular to the plural as is appropriate to the context or application. The various singular/plural permutations may be expressly set forth herein for sake of clarity. A reference to an element in the singular is not intended to mean "one and only one" unless specifically stated, but rather "one or more." Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the above description.

In general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as "open" terms (e.g., the term "including" should be interpreted as "including but not limited to," the term "having" should be interpreted as "having at least," the term "includes" should be interpreted as "includes but is not limited to," etc.). Furthermore, in those instances where a convention analogous to "at least one of A, B, and C, etc." is used, in general, such a construction is intended in the sense one having skill in the art would understand the convention (e.g., "a system having at least one of A, B, and C" would include but not be limited to systems that include A alone, B alone. C alone, A and B together. A and C together, B and C together, or A, B, and C together, etc.). Also, a phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to include one of the terms, either of the terms, or both terms. For example, the phrase "A or B" will be understood to include the possibilities of "A" or "B" or "A and B."

Additionally, the use of the terms "first," "second," "third," etc., are not necessarily used herein to connote a specific order or number of elements. Generally, the terms "first," "second." "third." etc., are used to distinguish between different elements as generic identifiers. Absence a showing that the terms "first," "second," "third," etc., connote a specific order, these terms should not be understood to connote a specific order. Furthermore, absence a showing that the terms first," "second," "third." etc., connote a specific number of elements, these terms should not be understood to connote a specific number of elements.

While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the scope of the invention. For example, the use of comprise, or variants such as comprises or comprising, includes a stated integer or group of integers but not the exclusion of any other integer or group of integers. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is intended that any claims presented at any time in this application define the scope of the invention and that methods and structures within the scope of these claims and their equivalents are covered thereby.

What is claimed is:

1. A surf pool, comprising:

a surf basin;

a wave generator disposed on a first side of the surf basin and configured to generate waves that travel from the first side of the surf basin to a second side of the surf basin opposite the first side of the surf basin; and

a return channel comprising a peripheral channel portion extending along at least a portion of a periphery of the second side of the surf basin and a central channel portion that is connected to and protrudes from the peripheral channel portion and into a central region of the surf basin, the return channel configured to capture water therein, the central channel portion extending only partially across the surf basin from the second side towards the first side to reintroduce the captured water back into the surf basin at a middle region thereof between the first side and the second side;

wherein movement of the water captured in the peripheral channel portion is limited to movement along the periphery of the surf basin until the water enters the central channel portion at which point the water is able to move directly toward the first side of the surf basin.

2. The surf pool of claim **1**, wherein a central channel width of the central channel portion is larger than a peripheral channel width of the peripheral channel portion.

3. The surf pool of claim **1**, further comprising a plurality of grating panels covering at least the central channel portion of the return channel such that upward facing surfaces of the plurality of grating panels are flush with and follow an upward slope of the surf basin.

4. The surf pool of claim **3**, further comprising a plurality of beams extending across a return channel width of the return channel and supporting the plurality of grating panels.

5. The surf pool of claim **3**, further comprising a plurality of concrete piers disposed within the return channel and operable to support the plurality of grating panels.

6. The surf pool of claim **4**, wherein the return channel width of the return channel is at least 4 meters.

7. The surf pool of claim **1**, further comprising an edge pool, wherein the edge pool shares a wall with the return channel.

8. The surf pool of claim **7**, further comprising a plurality of grating panels covering the return channel such that one or more of the plurality of grating panels are in direct contact with the wall shared by the return channel and the edge pool.

9. The surf pool of claim **1**, further comprising an edge pool disposed between and separating a first section of the return channel and a second section of the return channel.

10. The surf pool of claim **9**, wherein the return channel is configured to transfer water from the first section to the second section by way of the edge pool.

11. The surf pool of claim **9**, wherein the edge pool is in fluid communication with the first section and the second section of the return channel.

12. The surf pool of claim **1**, wherein the surf basin is defined by a plurality of side walls and a bottom surface and wherein the return channel comprises a channel formed in the bottom surface.

13. The surf pool of claim **1**, wherein the central channel portion is a first central channel portion and wherein the return channel further comprises a second central channel portion that is connected to and protrudes from the peripheral channel portion and into the central region of the surf basin.

14. A return channel, the return channel comprising:

a peripheral channel portion extending along at least a periphery of a first side of a surf basin, opposite a second side of the surf basin;

a central channel portion that is connected to and protrudes from the peripheral channel portion and into a central region of the surf basin, the return channel configured to capture water therein, the central channel portion configured to extend only partially across the surf basin from the first side towards the second side to reintroduce the captured water back into the surf basin at a middle region thereof between the first side and the second side; and

a plurality of grating panels covering at least a portion of the peripheral channel portion and the central channel portion,

wherein movement of water captured in the peripheral channel portion is limited to movement along the periphery of the surf basin until the water enters the central channel portion at which point the water is able to move toward the central region of the surf basin.

15. The return channel of claim **14**, wherein upward facing surfaces of the plurality of grating panels are flush with and follow an upward slope of the surf basin.

16. The return channel of claim **14**, wherein the central channel portion protrudes from the peripheral channel portion and toward a wave generator located on a second side of the surf basin opposite the first side of the surf basin.

17. The return channel of claim **14**, further comprising a plurality of beams extending across a return channel width of the return channel and supporting the plurality of grating panels.

18. The return channel of claim **14**, further comprising a plurality of concrete piers disposed within the return channel and operable to support the plurality of grating panels.

19. The return channel of claim **14**, further comprising an edge pool, wherein the edge pool shares a wall with the peripheral channel portion of the return channel.

20. The return channel of claim **14**, further comprising an edge pool disposed between and separating a first section of the peripheral channel portion and a second section of the peripheral channel portion.

21. The return channel of claim **20**, wherein the return channel is configured to transfer water from the first section to the second section by way of the edge pool.

22. The surf pool of claim **8**, wherein:

at least one grating panel of the plurality of grating panels comprises a plurality of openings; and

the plurality of openings define at least one of:

at least two differing shapes; or

at least two differing sizes.

23. The return channel of claim **14**, wherein:

at least one grating panel of the plurality of grating panels comprises a plurality of openings; and

the plurality of openings define at least one of:

at least two differing shapes; or

at least two differing sizes.

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